ROS observation process
Base/wrist RGB + joints + estimated gripper

localhost request

Separate model process
SmolVLA checkpoint → 50 × 7 actions

Demonstration references
Approach / carry / release locations

prediction

Phased executor
Select ≤5 actions; validate; replace queue
Nominal publication: 20 Hz

/control_abstraction_layer
Six absolute joints + binary gripper

Control engine
Target checks + filtering
Velocity/acceleration limits

125 Hz arm commands
Forward position controller → UR5

Discrete gripper SetIO
Two sequential writes

Acknowledged I/O state
Not measured grasp success